State density IV: the box form of the divisor-coefficient theorem

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Abstract

The chart standard integral over $(0,b]^{d+1}$ with general continuous ξ,η is identified, by Fubini at the last coordinate, with the state integral of unit 70. Consequently, when the last coordinate carries the strictly smallest candidate exponent and ξ,η are Lipschitz in it at 0 with $\eta>0$ on the divisor, $\int_{(0,b]^{d+1}} w^h\eta(w)e^{-\beta n w^{2k}+\beta\sqrt{n}\,w^k\xi(w)}\,dw \sim C(\xi,\eta)\,n^{-p/2}$ with the divisor coefficient of unit 71. Zero `sorry/axiom`.

1 The things formalised

```
noncomputable def boxIntegralGen (  b c :  ) {m :  } (k h : Fin m →  )
    (   : (Fin m →  ) →  ) :   :=
   w : Fin m →  , (  i, w i ^ h i) *   w
    * Real.exp (-  * (c *   i, w i ^ k i) ^ 2
        +   * (c *   i, w i ^ k i) *   w)  (boxMeasure b m)
theorem continuous_snoc_prod (d :  ) :
    Continuous fun x :   × (Fin d →  ) => (Fin.snoc x.2 x.1 : Fin (d + 1) →  )
theorem boxIntegralGen_eq_stateIntegral (  b n :  ) (hn : 0   n) ... :
    boxIntegralGen   b (Real.sqrt n) k h
      = stateIntegral   b n (h (Fin.last d)) (k (Fin.last d))
          (fun i => k (Fin.castSucc i)) (fun i => h (Fin.castSucc i))
          (fun u v =>   (Fin.snoc v u)) (fun u v =>   (Fin.snoc v u))
theorem boxIntegralGen_isEquivalent ... (hq : last exponent strictly minimal)
    (h  h  : Lipschitz in the last coordinate at 0) (h pos :   > 0 on the divisor) :
    (fun n => boxIntegralGen   b (Real.sqrt n) k h    ) ~[atTop]
      fun n => (  v, divisorDensity ... v  (boxMeasure b d))
        * n ^ (-(((h (Fin.last d) :  ) + 1) / k (Fin.last d) / 2))
```

2 Proof strategy

The bridge is the unit-54 idiom: `measurePreserving_piFinSuccAbove` at `Fin.last` turns the pi measure on $\mathbb{R}^{d+1}$ into the product of the last-coordinate measure with the pi measure on $\mathbb{R}^d$; `integral_prod_symm` then iterates with the last coordinate innermost. The pointwise identity $F(w)=g(e\,w)$ uses `Fin.snoc_init_self`, `Fin.prod_univ_castSucc` and $(\sqrt{n})^2=n$. Integrability of the transported integrand is `integrable_comp_emb` plus continuity of the box integrand. The corollary composes `stateIntegral_isEquivalent` with the bridge, eventually in $n\geq 0$.

3 Roadblocks and resolutions

- The coordinate identity $(e\,w).2\,j = w(j.\mathrm{castSucc})$ is `congrArg w (Fin.succAbove_last_apply j)`; rewriting fails because the goal already shows the unfolded form.
- A trailing $\partial\mu$ after a nested integral binds the inner integral: parenthesise the inner integral (same gotcha as unit 70).
- Instead of a case split on the sign of n inside `congr_left`, use `filter_upwards [eventually_ge_atTop 0]`: eventual equality suffices.

4 Outcome

Programme A (unique minimal coordinate, general ξ, η) is complete in the genuine box form. A GPT-6 Astra consult (`tide-log/gpt6_bigpicture_v1.md`) fixes the plan for multiplicity $m \geq 2$: exact product-density identity for the minimal block, fixed-strip freezing, and $C = \frac{1}{2^{m-1}(m-1)!\prod k_i} \int v^{h'-pk'} \eta_0 A_{p-1}(\xi_0)\, dv$ with $\log^{m-1} n$.